

IoT Smart Helmet System for Delivery Rider Safety (Helmify)

1. Problem Statement

The rapid growth of online shopping and food delivery services has significantly increased the number of delivery riders on the road. These riders face several safety challenges, such as riding without helmets, delayed accident detection, lack of real-time emergency assistance, and poor communication with delivery companies. Existing delivery systems mainly focus on order management but do not provide rider safety monitoring. As a result, accidents often go unnoticed for several minutes, increasing the risk of serious injuries or fatalities.

The objective of this project is to develop an IoT-based Smart Helmet System that enhances rider safety by monitoring helmet usage, detecting accidents in real time, providing live GPS tracking, and enabling communication between riders and administrators through a mobile application and web dashboard.

2. Problems Solved by the System

The proposed system addresses the following real-world problems:

- Ensures riders wear helmets before starting their journey.
 - Detects accidents automatically without requiring manual intervention.
 - Sends instant accident alerts for faster emergency response.
 - Provides real-time GPS location tracking of delivery riders.
 - Allows administrators to monitor rider safety and delivery status.
 - Improves communication between riders and delivery companies.
 - Reduces accident response time.
 - Increases overall delivery efficiency and operational transparency.
 - Promotes safe riding practices and helmet compliance.
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3. Key Features

Smart Helmet Monitoring

- Detects whether the rider is wearing the helmet using an FSR sensor.

Accident/Fall Detection

- Uses an MPU6050 accelerometer and gyroscope to detect sudden falls or accidents.

Real-Time Alerts

- Immediately sends accident notifications to the rider's mobile application and administrator dashboard.

Live GPS Tracking

- Tracks the rider's real-time location using the smartphone GPS.

Android Mobile Application

- Connects to the smart helmet via Bluetooth.
- Displays helmet status.
- Shows delivery orders.
- Provides Google Maps navigation.
- Displays accident notifications.
- Allows riders to accept or reject delivery requests.

Administrator Dashboard

- Displays live rider locations.
- Monitors helmet compliance.
- Receives accident alerts.
- Tracks delivery progress.
- Manages rider information and orders.

Bluetooth Communication

- Uses Bluetooth Low Energy (BLE) for communication between the helmet and smartphone.

Cloud Integration

- Stores rider information, order details, safety alerts, and live location using Firebase.

Hands-Free Communication

- Supports Bluetooth headset integration for safer communication while riding.
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4. Technology Stack

Hardware

- ESP32 Microcontroller
 - MPU6050 Accelerometer & Gyroscope
 - Force Sensitive Resistor (FSR)
 - Bluetooth Low Energy (BLE)
 - 6800mAh Li-ion Battery
 - TP4056 Charging Module
 - Bluetooth Headset (Microphone & Speaker)
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Software

Mobile Application

- Android Studio
- Java / Kotlin
- Google Maps API
- Firebase Authentication
- Firebase Firestore

Web Dashboard

- HTML
- CSS
- JavaScript
- Firebase
- Google Maps API

Embedded Programming

- Arduino IDE
 - C++
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Cloud Services

- Firebase Firestore
 - Firebase Authentication
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5. Components Used

Component	Purpose
ESP32	Main microcontroller that processes all sensor data
MPU6050	Detects rider movement, falls, and accidents
FSR Sensor	Detects whether the helmet is being worn
Bluetooth (BLE)	Wireless communication between helmet and smartphone
Smartphone	Provides GPS location and internet connectivity
Firebase	Stores data and synchronizes information
Google Maps API	Navigation and live rider tracking
Li-ion Battery	Powers the smart helmet electronics
TP4056 Module	Safe battery charging
Bluetooth Headset	Hands-free communication

6. System Workflow

1. Rider wears the smart helmet.
2. FSR sensor checks whether the helmet is properly worn.
3. ESP32 continuously collects sensor data.
4. MPU6050 monitors rider movement.
5. If an accident is detected, ESP32 sends the alert via BLE.
6. Android application receives the alert.
7. Mobile app uploads accident information and GPS location to Firebase.

8. Administrator dashboard displays the rider's live location and accident notification.
 9. Company can immediately contact the rider or dispatch emergency assistance.
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7. Advantages

- Improves rider safety.
 - Reduces accident response time.
 - Encourages helmet usage.
 - Enables real-time rider monitoring.
 - Cost-effective IoT solution.
 - Easy to deploy.
 - Scalable for large delivery fleets.
 - Improves delivery management.
 - User-friendly mobile application and dashboard.
 - Low power consumption using BLE.
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8. Limitations

Although the system provides significant safety improvements, it has some limitations:

- The helmet depends on the rider's smartphone for GPS and internet connectivity.
 - If the smartphone battery is discharged, live tracking and cloud communication stop.
 - Bluetooth communication range is limited.
 - False accident detection may occur during sudden movements.
 - Requires regular charging of the helmet battery.
 - Performance may be affected in areas with poor mobile network coverage.
 - Currently designed mainly for delivery riders and may require modifications for other applications.
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9. Future Scope

The system can be enhanced with advanced technologies to improve safety and functionality:

- AI and Machine Learning for more accurate accident detection.

- Drowsiness detection using camera or eye-tracking sensors.
 - Alcohol detection before starting the vehicle.
 - Automatic SOS messages to family members and hospitals.
 - Heart rate and body temperature monitoring.
 - Voice assistant integration for hands-free operation.
 - Predictive accident analysis using historical rider data.
 - Vehicle health monitoring.
 - Integration with smart city infrastructure.
 - Support for 5G and IoT edge computing.
 - Cloud analytics for rider performance and safety reports.
 - Fleet management dashboard with advanced analytics.
 - Integration with ambulance and emergency response systems.
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10. Expected Outcome

The proposed IoT Smart Helmet System enhances the safety of delivery riders by integrating smart sensors, IoT communication, mobile applications, and cloud services into a single platform. It helps reduce accident response time, ensures helmet compliance, enables real-time monitoring, and improves delivery operations. The system provides a practical, scalable, and cost-effective solution for modern delivery services while contributing to safer transportation and more efficient logistics.